CDS 101 -Final Project Report

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1. Problem Definition

- Our project investigates whether weather conditions are related to equine incidents at New York race tracks. The main question we are studying is: How significant is the relationship between equine incidents and weather conditions?
- This topic is important because equine deaths and injuries are serious safety concerns in horse racing. By studying incident patterns over time and comparing them with weather conditions, we may be able to identify whether certain weather factors are associated with higher or lower incident counts.
- The objective of this analysis is to explore trends in equine incidents from 2009 to 2025, compare incidents across different years, tracks, incident types, seasons, and weather categories, and determine whether weather appears to have a noticeable relationship with incident patterns.
- A key assumption is that the incident reports are accurate and consistently recorded. Another assumption is that the weather information connected to each incident is reliable enough to be grouped into categories such as clear, rainy, cloudy, snowy, or unknown.

2. Data Acquisition & Description

- The name of the dataset is Equine Death and Breakdown from the data.gov website which is an official website of the United States government.
- Obtained through a download link from data.gov website.
- There are a lot of variables / features such as Incident Type, Horse, Jockey Driver, Track, and so forth but we will be focusing primarily on two variables, Death or Injury and Weather Conditions. Death or Injury variable shows if a horse died in an incident or was just injured. Weather Conditions variable shows the weather conditions at the time of the incident. We will also be using "Track", "Year", and "Incident Date" columns to help supplement our findings.
- The dataset has 5,383 rows and 13 columns.

3. Data Cleaning & Preprocessing

- This dataset was extremely messy and we had to do a lot of cleaning to make it usable. We got rid of any rows that had NA values in the Weather Condition and Death and Injury variables. The incident date variable had to be changed to a date type. The column names were fixed to make it easier to use and read. Originally the dataset's weather condition variable had 1000+ different unique entries but we wanted them to be categorized in just a few simple ones. The weather_clean variable that was created that holds the weather_conditions values without all the unnecessary information and spaces. This is then piped into weather_group which categorizes the specific weather condition happening at the time of the incident such as "Rain", "Snow", or "Clear" and so forth. The death_or_injury_clean variable cleans up the 10 unique values and categorizes them into 2 simple ones "death" or "injury." After creating these new variables we filtered them to only include the values that were relevant to our research question. A season variable was also created for EDA. Finally, the Weather Condition variable was cleaned so that way The Death and Injury variable had to be encoded to binary 0/1 variable since its a categorical variable and we need a numerical values to make a linear regression model.

```
#Fix column names
horses_clean <- horses %>%
  rename(year = Year, incident_date = `Incident Date`, track = Track, inv_location = `Inv Locat

#Fix dates
horses_clean$incident_date <- mdy(horses_clean$incident_date)

#Get rid of NA entries
horses_clean <- horses_clean %>%
  filter(!is.na(weather_conditions), !is.na(death_or_injury))

#Fix weather_condition column
#USED CHATGPT FOR THIS CODE!!!!!!!!!

horses_clean <- horses_clean %>%
  mutate(
    weather_clean = weather_conditions %>%
      str_to_lower() %>%
      str_trim() %>%
      str_replace_all("[0-9]+", "") %>%        # remove numbers
      str_replace_all("[^a-z ]", "") %>%       # remove symbols
      str_replace_all("couldy|cloudyy|clody", "cloudy") %>%
      str_replace_all("raine", "rain") %>%
      str_squish()
  ) %>%

  mutate(
    weather_group = case_when(
```

```r
      # Rain
      str_detect(weather_clean, "rain|shower|drizzle") ~ "Rain",

      # Snow
      str_detect(weather_clean, "snow|flurr|sleet") ~ "Snow",

      # Storms
      str_detect(weather_clean, "thunder|storm") ~ "Storm",

      # Cloudy / Overcast
      str_detect(weather_clean, "cloud|coudy|cloudy") ~ "Cloudy",

      # Clear / Sunny
      str_detect(weather_clean, "clear|sun|fair|dry") ~ "Clear",

      # Other weather conditions
      str_detect(weather_clean, "windy|fog|mist|hazy|wind") ~ "Other Weather",

      # Missing / junk
      weather_clean %in% c("", "na", "n a", "unc") | is.na(weather_clean) ~ "Unknown",

      TRUE ~ "Other"
    )
  )

#Clean death_or_injury column
horses_clean <- horses_clean %>%
  mutate(
    death_or_injury_clean = death_or_injury %>%
      str_to_lower() %>%
      str_trim(),

    death_or_injury_clean = case_when(
      str_detect(death_or_injury_clean, "death|dead|euthanasia") ~ "Death",
      str_detect(death_or_injury_clean, "lame|injury|lameness") ~ "Injury",
      TRUE ~ "Other"
    )
  )

horses_clean <- horses_clean %>%
  filter(death_or_injury_clean != "Other") %>%
  select(year, incident_date, incident_type, track, incident_description, weather_group, death_
  filter(weather_group != "Other" & weather_group != "Unknown")

#Create season variable (Code reference linked below)
horses_clean <- horses_clean %>%
  mutate(
```

```r
    season = case_when(
    month(incident_date) %in% 9:11 ~ "Fall",
    month(incident_date) %in% c(12,1,2) ~ "Winter",
    month(incident_date) %in% 3:5 ~ "Spring",
    TRUE ~ "Summer"
  ))
```

4. Exploratory Data Analysis (EDA)

This section maps directly to the **EDA** part of the rubric:

- Summary statistics
- Visualizations
- Interpretation connected to the research question

```r
summary(horses_clean)
```

```
##       year      incident_date        incident_type         track
##  Min.   :2009   Min.   :2009-04-28   Length:1702        Length:1702
##  1st Qu.:2012   1st Qu.:2012-01-20   Class :character   Class :character
##  Median :2016   Median :2016-10-01   Mode  :character   Mode  :character
##  Mean   :2017   Mean   :2017-03-27
##  3rd Qu.:2022   3rd Qu.:2022-09-15
##  Max.   :2026   Max.   :2026-04-12
##  incident_description weather_group      death_or_injury_clean
##  Length:1702          Length:1702        Length:1702
##  Class :character     Class :character   Class :character
##  Mode  :character     Mode  :character   Mode  :character
##
##
##
##     season
##  Length:1702
##  Class :character
##  Mode  :character
##
##
##
```

```r
library(ggplot2)
ggplot(horses, aes(x = Year, color = 'white')) +
geom_point(stat = "count", size = 3) +
```

```r
labs(
  x = "Year",
  y = "# of Incidents",
  title = "Equine Incidents Over Time"
  ) +
theme_minimal()
```

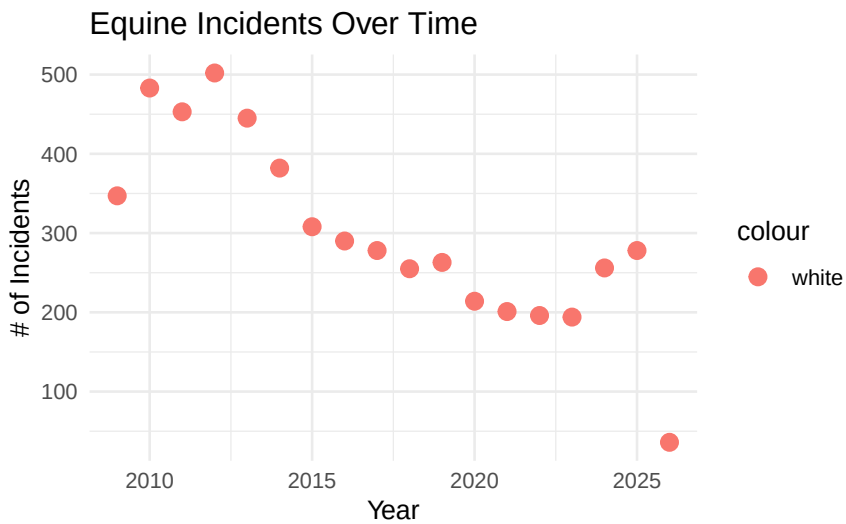


Figure 1: Relationship between time of year and weather related equine incidents

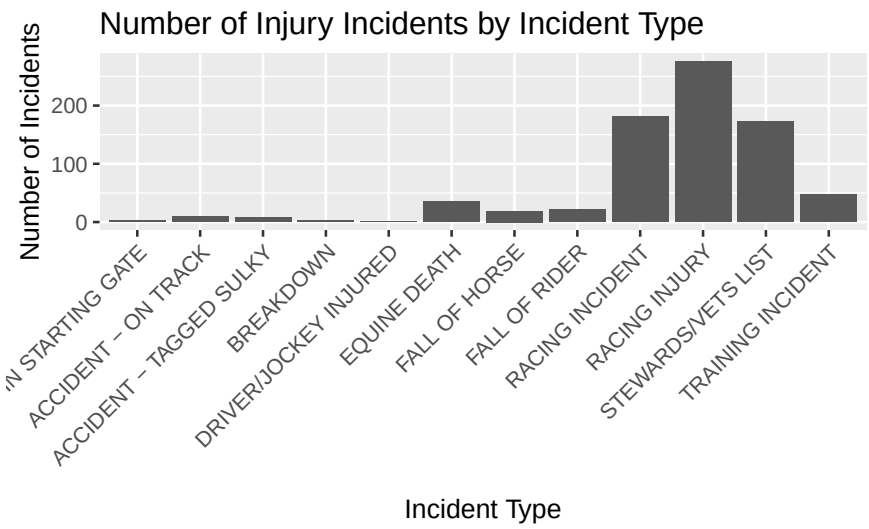
```r
library(readr)

Equine_Death_and_Breakdown <- read_csv("data/Equine_Death_and_Breakdown.csv")
```

```
## Rows: 5381 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): Incident Date, Incident Type, Track, Inv Location, Racing Type Des...
## dbl  (1): Year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
horses_clean %>%
  filter(death_or_injury_clean == "Injury") %>%
  ggplot(aes(x = incident_type)) +
  geom_bar() +
  labs(
    title = "Number of Injury Incidents by Incident Type",
    x = "Incident Type",
```

```
    y = "Number of Incidents"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



- This graph connects to our research question by showing which types of equine incidents happen most often in the data set. Since our project is studying the relationship between weather related variables and equine deaths or breakdowns, we first need to understand what kinds of incidents are being recorded. The most common incident types are the main categories we can later compare with weather conditions, temperature, precipitation, or season. This graph does not show a weather relationship by itself, but it gives important background for the weather analysis.

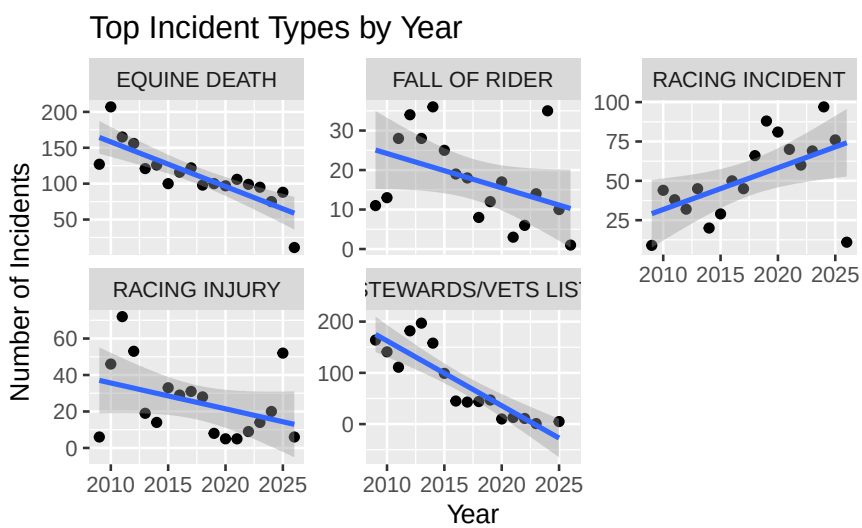
```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats 1.0.1     v tibble  3.3.1
## v purrr   1.2.1     v tidyr   1.3.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
Equine_Death_and_Breakdown %>%
  filter(`Incident Type` %in% c(
    "EQUINE DEATH",
    "STEWARDS/VETS LIST",
    "RACING INCIDENT",
    "RACING INJURY",
    "FALL OF RIDER"
```

```
  )) %>%
  count(Year, `Incident Type`) %>%
  ggplot(aes(x = Year, y = n)) +
  geom_point() +
  geom_smooth(method = "lm") +
  facet_wrap(~ `Incident Type`, scales = "free_y") +
  labs(
    title = "Top Incident Types by Year",
    x = "Year",
    y = "Number of Incidents"
  )
```

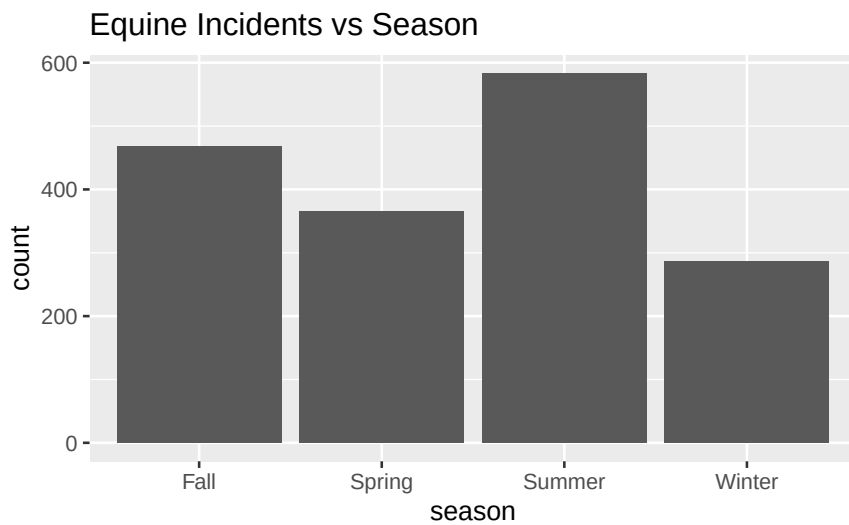
```
## `geom_smooth()` using formula = 'y ~ x'
```



- These graphs help connect to our question by showing how incident types changed over time. If certain years had higher numbers of specific incident categories, those changes could later be compared with weather patterns from those same years. This helps us look for possible trends before testing whether weather-related variables are connected to equine deaths or breakdowns. The graph does not prove that weather caused the incidents, but it helps identify time-based patterns that may be useful for the rest of the project.

ii.

```
horses_clean %>%
  ggplot(aes(x = season)) +
  geom_bar() +
  labs(
    title = "Equine Incidents vs Season"
  )
```



- We assume that Winter has the most inclement weather out of all the seasons and yet we see that Winter has the least amount of Equine deaths and injuries which could indicate a weak correlation between season and Equine incidences if we take our assumption as the truth.

iii.

```
#Weather vs Incidents Graph
horses_clean %>%
ggplot(aes(x = death_or_injury_clean, fill = death_or_injury_clean)) +
  geom_bar() +
  facet_wrap(~ weather_group) +
  labs(
    x = "Death or Injury",
    title = "Equine Incidents vs Weather Condition")
```

- There does not appear to be a clear correlation between weather condition and Equine death or injury.

5. Visualization Quality and Storytelling

Use this section to satisfy the **Visualization Quality** rubric criterion:

- Explain why your plot types are appropriate.
- Comment on labels, legends, colors, and overall readability.
- Mention any steps you took to make plots accessible and interpretable.

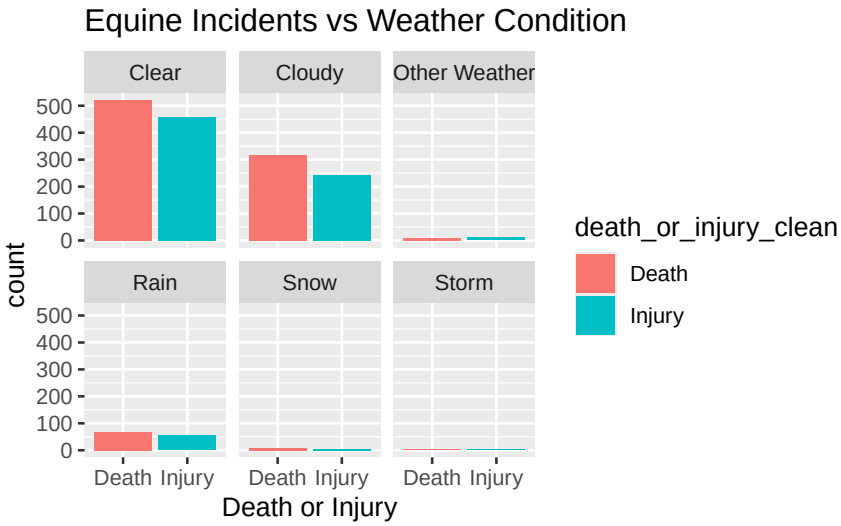


Figure 2: Example histogram of a numeric variable (EDA rubric).

- For the incident type section, we used a bar graph first because we are looking at incident type which is categorical.This graph makes sense because it lets us compare how many incidents happened in each incident type, and we flipped the graph so the incident type names are easier to read instead of crowded on the x-axis. For the second graph, we used a line graph because we are looking at incident type by year, which helps show how the incident categories changed over time instead of only showing the total number of incidents.
- For graphs ii. and iii. we used bar graphs because we were comparing two categorical variables to each other.

6. Modeling Approach

Explain how you framed the problem and which models you chose:

- Type of task (regression, classification, etc.).
- Baseline model or heuristic, if used.
- Main model(s) chosen and why they are appropriate.

For our modeling approach, we followed the plan from our proposal by using ideas from the blood pressure assignment and the car prices assignment. In the blood pressure assignment, we looked at relationships between variables and used linear models to understand trends. In the car prices assignment, we used a model with more than one explanatory variable. For our project, we used the number of incidents as the response variable because incident count is numeric and works with a linear model. The explanatory variables we focused on were year, season, weather condition, and incident type. These variables match the graphs we made because we looked at incidents over time, incident categories, incidents by season, and incidents by weather condition.

For our baseline model, we would use a simple heuristic, which is a simple rule for making a prediction. Our heuristic would predict the average number of incidents every time without using year, season, weather condition, or incident type. This gives us a simple comparison for our multi-variable linear model. If the multi-variable model makes better predictions than just using the average incident count, then that would suggest that the variables we chose are useful for explaining patterns in the data. This connects to our research question because we are trying to see whether weather-related variables are connected to equine deaths and breakdowns. The incidents-over-time graph shows that incident counts changed by year, so year is an important variable to include. The incident type graphs show which categories happened the most and whether those categories changed over time. The season and weather condition graphs connect more directly to the weather part of our project because they show how incidents are spread across different seasons and weather conditions. Using a multi-variable linear model helps us look at these variables together instead of only looking at one graph or one factor at a time.

7. Model Implementation & Evaluation

This corresponds to the **Model Implementation & Evaluation** rubric criterion.

Describe:

- Features used.
- Data splitting strategy (train/test or cross-validation).
- Metrics used (accuracy, F1, RMSE, etc.).
- Tables/plots summarizing performance.
- A short interpretation of each metric/plot.

```
# Example structure:
# set.seed(123)
# train_index <- sample(seq_len(nrow(data_clean)), size = 0.8 * nrow(data_clean))
# train <- data_clean[train_index, ]
# test  <- data_clean[-train_index, ]
#
# model <- glm(target ~ ., data = train, family = binomial)
# preds <- predict(model, newdata = test, type = "response")
# # compute metrics...
```

For the model implementation, the response variable would be the count of incidents, and the main features would be year, season, weather condition, and incident type. We included year as a time trend variable because the incidents-over-time graph showed that incident counts were not the same every year. This matters because the model should account for changes over time instead of treating all years as equal. We included season and weather condition because our research question focuses on weather-related factors. We also included incident type because different categories of incidents may have different patterns.

The graphs support these choices because they show that incidents are not evenly spread across the dataset. The incident type graph shows that some incident categories happen more often than others. The incident type by year graph shows that those categories also changed over time. The season and weather condition graphs show that incidents were different across seasons and weather conditions. Clear and cloudy weather had higher recorded incident counts, but we have to be careful with that interpretation because it does not automatically mean those conditions caused more incidents. There may have been more races during clear or cloudy weather, which would create more chances for incidents to happen.

To evaluate the model, we would compare the model's predicted incident counts to the actual incident counts. We would also compare the model to our baseline heuristic, which only predicts the average number of incidents. If the multi-variable model gives predictions that are closer to the actual incident counts than the simple average, then the variables we chose are helping explain the data. Overall, this follows our proposed method because we used time trends, incident categories, weather-related variables, and a multi-variable linear model to look for patterns connected to equine deaths and breakdowns.

8. Conclusions & Recommendations

Summarize the key takeaways:

- Answer your original research question(s) directly.
- Highlight the most important patterns or relationships you found.
- Discuss limitations (data size, bias, missing variables, etc.).
- Suggest possible extensions or future work.
- Possible

9. Code Quality & Reproducibility

Briefly document how someone else can reproduce your results:

- Which R scripts or Rmd files to run.
- Any required R packages.

-Explain how code is structured and include a link to your Git repository

10. References

List any references you used, such as:

- Dataset documentation.
- Research papers or articles.
- Tutorials or blog posts.
- Dates to seasons column tutorial